

Activity 2 – Report in KNIME

Creating Analytical Reports Using KNIME Analytics Platform

Aim

The aim of this activity is to design and generate a professional report within KNIME Analytics Platform by combining data tables, visualizations, and layout elements using the KNIME Report Designer, thereby learning how to communicate analytical results in a structured document format.

Introduction

KNIME reporting is a feature of the KNIME Analytics Platform that allows users to compile the results of a data analysis workflow — such as tables, charts, and images — into a single, well-formatted document. Reports are used to present analytical findings in a clear, shareable, and reusable format such as PDF, HTML, or DOCX, making them suitable for stakeholders who may not directly interact with the workflow itself. The importance of reporting in data analytics lies in its ability to translate raw computational output into meaningful, presentable insights that support decision-making. Using the Report Designer extension, KNIME enables drag-and-drop placement of visualization and data components onto a structured page layout, combining headers, titles, tables, and charts. This bridges the gap between data processing and effective communication of results.

Procedure

The following steps outline the process of creating a report in KNIME, from setting up the workflow to generating the final document:

Step 1: Open KNIME Analytics Platform — Launch the KNIME Analytics Platform and create a new workflow in the desired workspace.

Step 2: Load the Dataset — Add a File Reader or CSV Reader node to import the dataset that will be used for the report.

Step 3: Process and Visualize the Data — Connect processing nodes as needed, then add visualization nodes such as Table View, Bar Chart, or Pie Chart to generate visual output from the data.

Step 4: Install/Open the Report Designer — Switch to the Report Design perspective in KNIME, which provides a canvas for arranging report elements such as titles, tables, and images.

Step 5: Design the Report Layout — Drag and drop the desired visualization and data output nodes onto the report canvas, and arrange headers, footers, and sections as required.

Step 6: Generate the Report — Execute the report generation process and export the final report in the desired format, such as PDF or DOCX.

Step 7: Review the Output — Open the generated report file to verify that all tables, charts, and layout elements appear correctly.

KNIME Workflow

The complete workflow designed for this activity connects the data-loading node, visualization nodes, and the Report Designer component. The overall structure of the workflow is shown below.

[Insert Complete KNIME Workflow Screenshot Here]

Figure 1: Complete KNIME workflow used for report generation

This screenshot should show the full node sequence — from data import through visualization to the Report Designer node — as connected on the KNIME workflow canvas.

Workflow Screenshots

This section documents each stage of the reporting workflow with supporting screenshots taken during the activity.

1. KNIME Analytics Platform

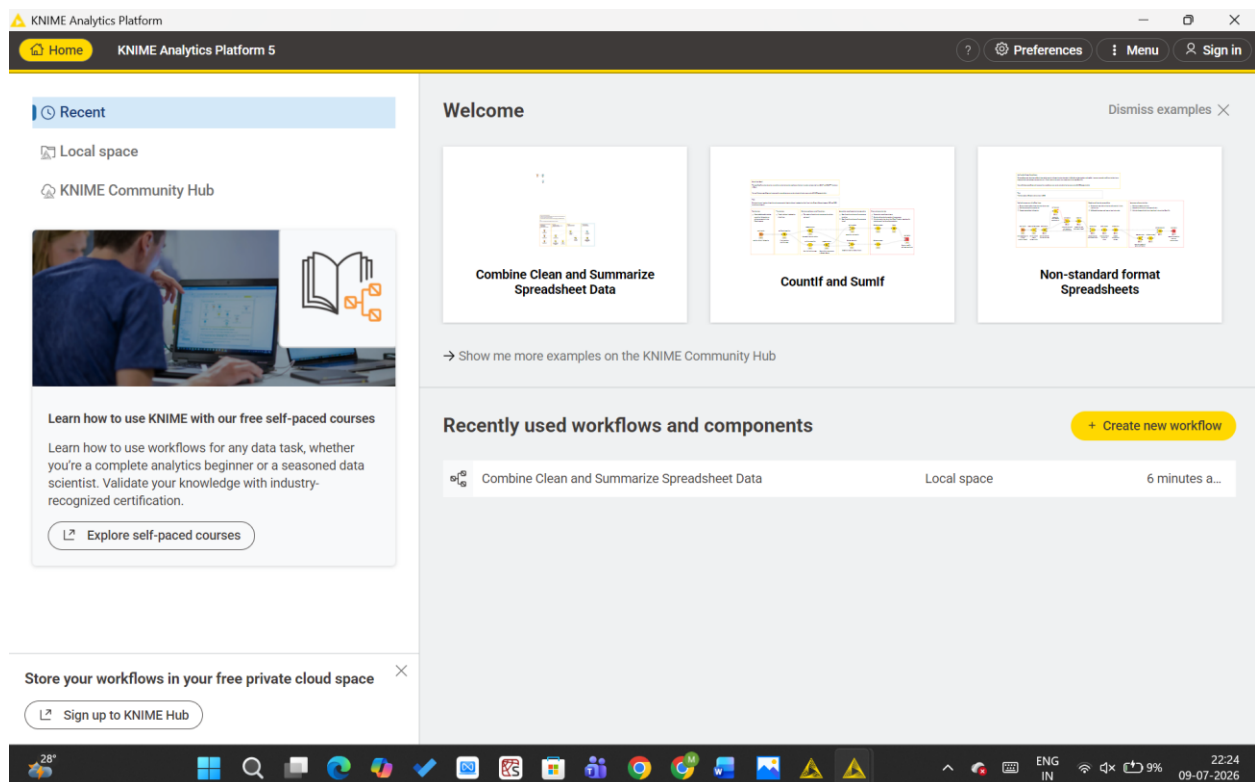


Figure 2: KNIME Analytics Platform home/workbench screen

Shows the KNIME Analytics Platform interface after launch, including the workflow repository, node repository, and workspace canvas.

2. Workflow Creation

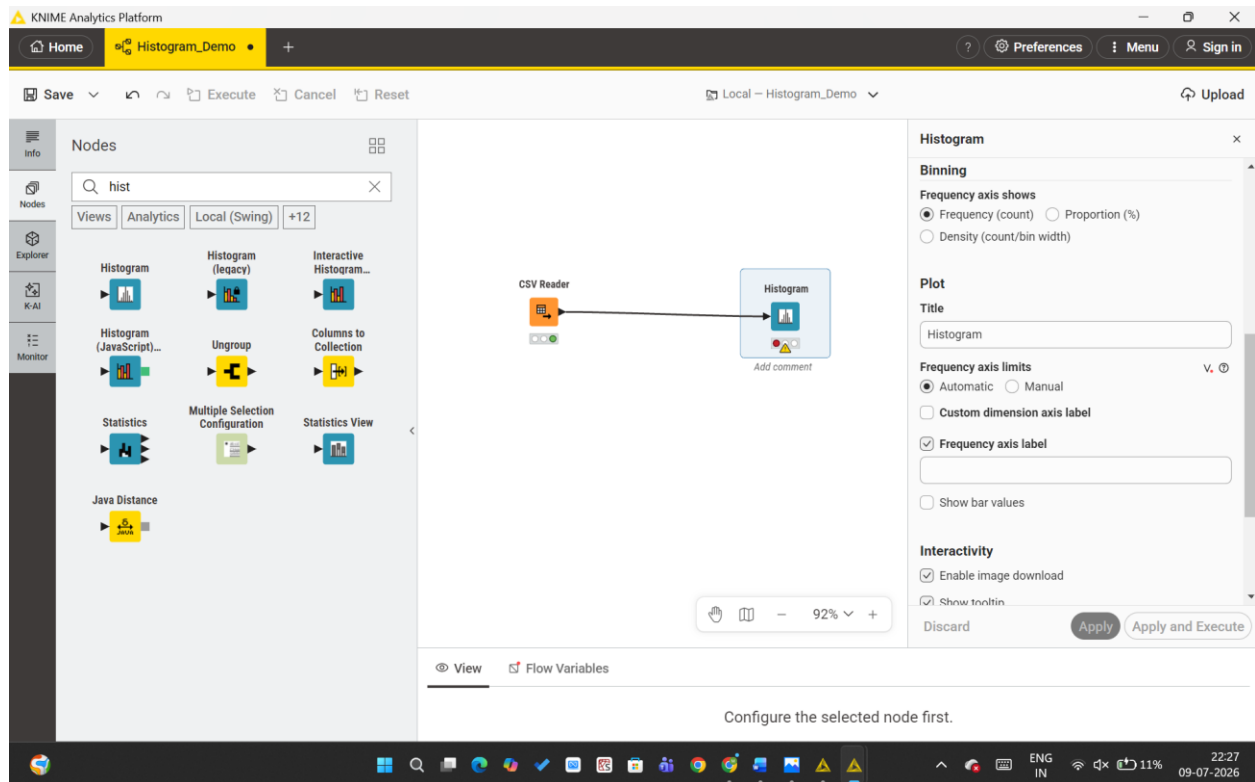


Figure 3: New workflow being created in the workspace

Displays the creation of a new, empty workflow in KNIME before any nodes have been added.

3. Data Loading

► 1: File Table ⚙ Flow Variables

Rows: 150 | Columns: 5 📊 Sorting 🔍 Filter Table Statistics 🔍 Search

☐ #	RowID	sepal length (cm) 📊 Number (Float)	sepal width (cm) 📊 Number (Float)	petal length (cm) 📊 Number (Float)	petal width (cm) 📊 Number (Float)	target 📄 String
☐ 1	Row0	5.1	3.5	1.4	0.2	Iris-setosa
☐ 2	Row1	4.9	3	1.4	0.2	Iris-setosa
☐ 3	Row2	4.7	3.2	1.3	0.2	Iris-setosa
☐ 4	Row3	4.6	3.1	1.5	0.2	Iris-setosa
☐ 5	Row4	5	3.6	1.4	0.2	Iris-setosa
☐ 6	Row5	5.4	3.9	1.7	0.4	Iris-setosa
☐ 7	Row6	4.6	3.4	1.4	0.3	Iris-setosa
☐ 8	Row7	5	3.4	1.5	0.2	Iris-setosa
☐ 9	Row8	4.4	2.9	1.4	0.2	Iris-setosa
☐ 10	Row9	4.9	3.1	1.5	0.1	Iris-setosa
☐ 11	Row10	5.4	3.7	1.5	0.2	Iris-setosa
☐ 12	Row11	4.8	3.4	1.6	0.2	Iris-setosa
☐ 13	Row12	4.8	3	1.4	0.1	Iris-setosa
☐ 14	Row13	4.3	3	1.1	0.1	Iris-setosa
☐ 15	Row14	5.8	4	1.2	0.2	Iris-setosa

Figure 4: Dataset being loaded using the CSV/File Reader node

Shows the configuration and successful execution of the data-reading node, confirming the dataset has been imported into KNIME.

4. Data Visualization

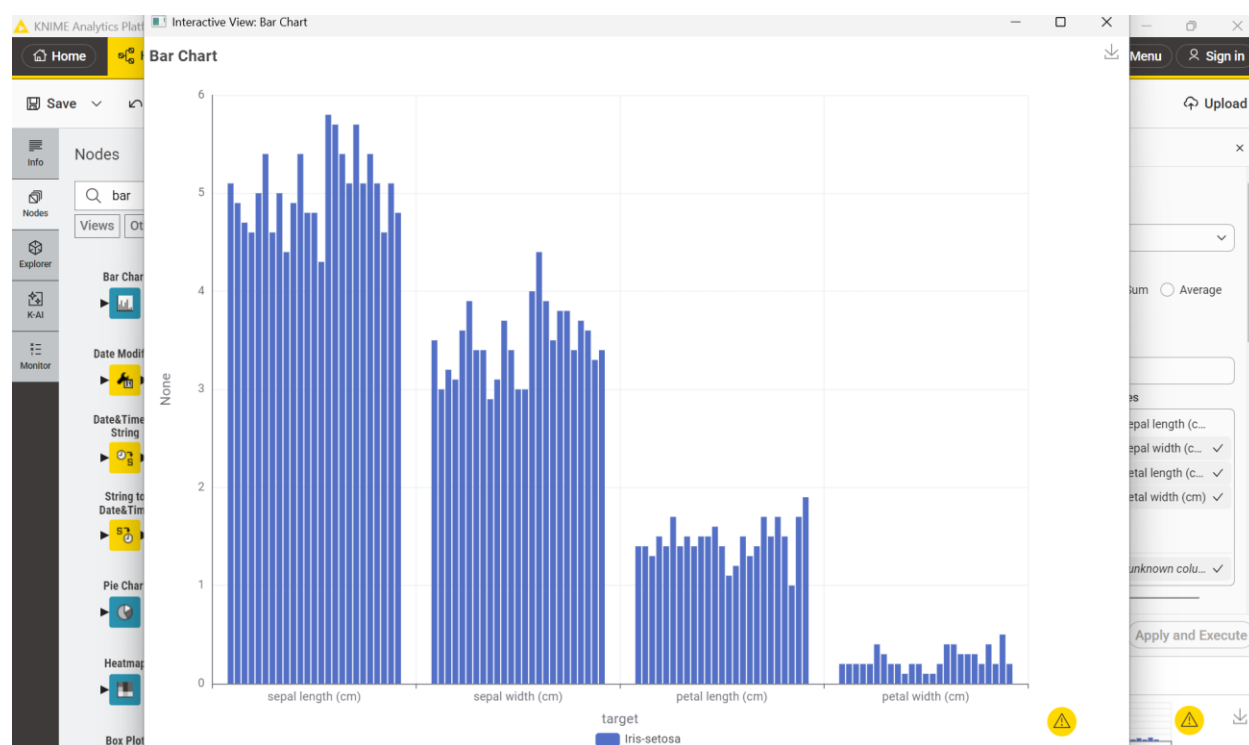


Figure 5: Visualization node output used in the report

KNIME Nodes Used

The table below briefly describes the purpose of each node used in the workflow:

Node	Description
File Reader / CSV Reader	Imports the dataset from a local file into the KNIME workflow as a data table for further processing.
Table View	Displays the loaded dataset in a row-and-column format, allowing quick inspection of the raw data.
Bar Chart / Pie Chart	Generates visual summaries of categorical data, which are later embedded into the report layout.
Column Filter	Selects only the relevant columns needed for the report, removing unnecessary data before visualization.
Report Designer (Layout Node)	Provides the canvas used to arrange tables, images, and charts into a structured report page.
Image / Report Output Node	Captures the output of visualization nodes as embeddable images for placement inside the report.
Report Export Node (PDF/DOCX)	Converts the final report layout into an exportable file format such as PDF or DOCX.

Generated Report

The final report generated through the KNIME Report Designer combines the loaded dataset, its visualizations, and layout formatting into a single output document. Sample pages of the generated report are shown below.

[Insert Generated Report Page 1 Here]

Figure 8: Page 1 of the final generated report

[Insert Generated Report Page 2 Here]

Figure 9: Page 2 of the final generated report

Result

A report was successfully created and generated in KNIME by combining the loaded dataset with corresponding visualizations using the Report Designer. The final exported document presented the data and charts in a clean, organized layout, confirming that the reporting workflow was configured and executed correctly.

Conclusion

Reporting is a vital part of the data analytics process, as it converts analytical results into a format that is easy to interpret, share, and present to others. KNIME's Report Designer simplifies this process by allowing users to visually assemble tables, charts, and text into a polished document without needing external tools. This activity demonstrated how workflow outputs can be transformed into professional reports suitable for business or academic communication. Mastering this feature enables analysts to deliver end-to-end solutions, from raw data processing to final presentation, all within a single platform.